



Targeting AgrA quorum sensing regulator by bumetanide attenuates virulence in *Staphylococcus aureus* – A drug repurposing approach

Balamurugan Palaniappan^{a,*}, Adline Princy Solomon^a, David Raj C^b

^a Quorum Sensing Laboratory, School of Chemical and Biotechnology, SASTRA Deemed University, Thanjavur 613401, Tamil Nadu, India

^b Central Animal Facility, School of Chemical and Biotechnology, SASTRA Deemed University, Thanjavur 613401, Tamil Nadu, India

ARTICLE INFO

Keywords:

Staphylococcus
Quorum sensing
Resistance
Antivirulence
Drug repositioning

ABSTRACT

Aims: The present study aims to target the quorum sensing (QS) accessory gene regulator A (AgrA) of *Staphylococcus aureus* to curtail bacterial virulence through drug repurposing approach.

Main methods: *In silico* screening of chemical ligands that bind specifically to the *S. aureus* C-LytTR domain of AgrA (AgrA_C) was carried out. AgrA inhibition and downregulation of virulence genes linked to QS system of *S. aureus* were determined. Efficacy, dermal toxicity and drug tolerance induction were tested in Balb/C mice dermonecrosis model.

Key findings: Bumetanide bound to the conserved amino acid Tyr-229 of AgrA and showed 70% AgrA inhibition at 0.1 μM. Highly significant reduction in the expression of representative virulence genes such as alpha-hemolysin (~5 log₂-fold), phenol-soluble modulins (~4 log₂-fold) and panton-valentine leukocidin (~3 log₂-fold) was noted *in vitro*. *In vivo* studies signified bumetanide to be highly effective in controlling the ulcer development and promoted wound healing. Also, the tested substance did not have dermal toxicity and no tolerance induction as well.

Significance: Targeting the QS regulators could be a possible alternative approach to curtail virulence in *S. aureus*. In addition, if the QS inhibitors are repurposed it could accelerate the drug development process and reduce the cost. The identified drug bumetanide inhibited AgrA and the results were in comparable to that of a known virulence inhibitor, diflunisal. The newly reported results of bumetanide in this study are expected to mark the drug's visibility for antibiotic adjunctive therapy and topical drug formulations for skin infections research.

1. Introduction

Staphylococcus aureus continues to be the predominant organism in purulent skin and soft tissue infections (SSTIs) [1]. Notably, methicillin-resistant *S. aureus* (MRSA) is more pronounced in healthcare associated infections worldwide [2,3]. Drug resistance has been reported to antibiotics like vancomycin, daptomycin, linezolid, ceftaroline, trimethoprim *etc.*, [1,4,5] and even to the current clinically prescribed topical antibiotic agent like mupirocin [6]. The wide and rapid emergence of multidrug resistance (MDR) in *S. aureus* has emphasized the need for alternative therapeutic strategies and drugs targeting them [7,8]. Quorum sensing (QS) based antivirulence approach is one among them and is highly promising with the report of various classes of QS inhibitors [9–13]. In *Staphylococci*, the accessory gene regulator (Agr) QS pathway is a more specific pathway that leads to the build up of the signal molecules as well as the expression of virulence factors like cell

wall associated proteins and exotoxins [14]. Increased Agr activity has been shown to decrease the biofilm and aid in biofilm dispersal [15–17]. Conversely, increased Agr expression results in the increased production several virulence factors. A few important of them that play a key role in *S. aureus* pathogenesis are the alpha-hemolysin (Hla) which is a pore forming toxin, phenol-soluble modulins (PSMα) that has multiple roles in staphylococcal pathogenesis and a component of panton-valentine leukocidin (lukS-PV) a cytotoxin that causes leukocyte destruction [18]. Several reports have shown that inhibiting AgrA significantly downregulates these virulence traits *in vitro* as well as *in vivo* [14,19–21]. Thus, inhibiting AgrA would be a practical approach to inhibit/reduce virulence, making the pathogen more vulnerable to the host immune system and antibiotic therapy. In addition, the QS inhibitor is highly significant, provided it to be an FDA approved drug, as it could readily adjuvant an existing antibiotic therapy [22,23]. In this regard, targeting the conserved C-terminal end of the LytTR domain of AgrA

* Corresponding author.

E-mail addresses: balamurugan@scbt.sastra.edu (B. Palaniappan), adlineprinzy@biotech.sastra.edu (A.P. Solomon).

<https://doi.org/10.1016/j.lfs.2021.119306>

Received 17 December 2020; Received in revised form 23 February 2021; Accepted 24 February 2021

Available online 1 March 2021

0024-3205/© 2021 Elsevier Inc. All rights reserved.

(AgrA_C), in *S. aureus* is highly warranted [24]. This site is known to be important for DNA-binding in the P2-P3 promoter region of the *Agr* operon that codes for quorum signals and virulence factors. Some of the virulence factors notable to mention are the enzymes involved in invasion, toxins, factors responsible for attachment, factors involved in persistence and evading host defences [25]. With an insight to this background, in the present study, we have aimed to identify FDA approved drugs that bind to *S. aureus* AgrA_C through molecular docking. Selected drugs were then tested *in vitro* for biofilm formation and antivirulence activities. Additionally, through Balb/C mice dermonecrosis model the efficacy, dermal toxicity and possibility of tolerance induction/resistance development to the test compound were also addressed. The promising results of the studies suggest that the reported drug could be a considerable option in formulating antibiotic adjunctive therapeutic research.

2. Materials and methods

2.1. *In silico*

2.1.1. Screening of FDA approved drugs and docking to C-LytTR domain of AgrA

The C-LytTR domain of AgrA (AgrA_C) of *S. aureus* (PDB 3BS1) was processed in PyMOL to remove the bound double-stranded DNA and magnesium ion. Ligands such as (4-phenoxyphenol, 9H-xanthene-9-carboxylic acid, 2-(4-methylphenyl)-1,3-thiazole-4-carboxylic acid and savirin (3-(4-propan-2-ylphenyl) sulfonyl-1H-triazolo[1,5-a]quinazolin-5-one) having *in vitro* inhibitory activity to AgrA of *S. aureus* were selected for pharmacophore generation [14,26]. The selected ligands were submitted to PharmaGist server to generate pharmacophores [27]. Of those, a savirin drug based pharmacophore with a score of 18.371 was selected for further screening of ligands to AgrA_C of *S. aureus*. The pharmacophore was submitted to zincpharmer FDA approved drug database to search for ligands. Ligands that qualified Lipinski's Rule of Five by QikProp module were energy minimized using LigPrep module of Schrodinger for further analysis. Molecular docking of the retrieved ligands binding to AgrA_C was performed using PyRx Autodock Vina [28,29]. Ligands that fit in to the pocket of transcriptional activation site of AgrA such as Leu-171, Glu-181 or His-200 and H-bonding to the conserved amino acid Tyr-229 were only scored [24]. The resulted drug molecules were considered for further *in vitro* studies.

2.2. *In vitro* assays

2.2.1. Bacterial strains and culture conditions

A standard *S. aureus* strain ATCC 25923 and a multidrug resistant clinical isolate (*S. aureus* AB459) were used to test the efficacy of the drugs *in vitro*. The drug resistance profile of the clinical strain *S. aureus* AB459 was detailed in an earlier study [13]. Laboratory strains *S. aureus* RN6390 (positive for *agr*) and *S. aureus* RN6911 Δ agr::tet(M) (*agr* null derivative of *S. aureus* RN6390) generously gifted by Professor Christiane Wolz (University of Tübingen) were chosen for internalization experiments. *S. aureus* RN6390B (pRN6683) was used for AgrA inhibition reporter assay [30]. All the strains were cultured in the differential medium mannitol salt agar (MSA) [HiMedia Laboratories, India]. Pure cultures of the strains were maintained as glycerol stocks at -80°C . All other assays were carried out in Tryptic Soy broth (TSB) supplemented with 0.5% glucose and incubated at 37°C .

2.2.2. Reference and test substances

The drug substances axitinib, bumetanide, dantrolene sodium salt and diflunisal were purchased from Sigma-Aldrich, India. All the test substances are hydrophobic and dissolved in cell culture grade DMSO. Initial stocks of the compounds were made in 100% DMSO and working concentrations were prepared by diluting with cell culture grade $1\times$ PBS or TSB media wherever necessary (maintaining final concentration of

DMSO to be 0.3%).

2.2.3. agrP3::blaZ reporter assay

All the compounds were tested for *Agr* activity by the *agrP3::blaZ* reporter assay. Of them, diflunisal was chosen as a reference compound in this study for two reasons; (i) in the present *in silico* study the compound resulted as a hit and capable of binding to AgrA_C (ii) in earlier studies, antivirulence property of diflunisal against *S. aureus* have been reported *in vitro* and *in vivo* [31,32]. This assay indirectly measures RNAPIII production by assessing the β -Lactamase activity [30]. β -Lactamase produced by the *S. aureus* *agrP3-blaZ* strain cleaves the chromogenic substrate nitrocefin to produce a red colour product and the increase in OD is measured. Briefly, *S. aureus* RN6390B (pRN6683) was grown overnight at 37°C in TSB medium containing $5\text{ }\mu\text{g/mL}$ chloramphenicol. The culture was then diluted 1:100 in fresh TSB medium and incubated till log phase (OD_{600} 0.4) was determined as detailed earlier. Briefly, $45\text{ }\mu\text{L}$ of the log phase culture was incubated with $5\text{ }\mu\text{L}$ of the test compounds, at varying concentrations ($100\text{--}0.015\text{ }\mu\text{M}$). For positive control, instead of the drugs, $5\text{ }\mu\text{L}$ of the cell-free supernatant from the overnight culture was used. The plates were incubated at 37°C for 60 min. $50\text{ }\mu\text{L}$ of sodium azide solution (5 mM) in TSB media was used to stop the reaction. Then, nitrocefin, $50\text{ }\mu\text{L}$ ($125\text{ }\mu\text{g/mL}$ solution) was added to determine the β -Lactamase activity. 1 unit of β -Lactamase activity was defined as the increase in 0.001 OD units per minute and the readings were taken in a multiwell plate reader in kinetic mode at $490\text{--}650\text{ nm}$ for 120 min.

2.2.4. Virulence gene expression determination by qRT-PCR

Quantitative RT-PCR was carried out to determine the virulence gene expressions (representative virulence genes as shown in Table S1) in *S. aureus* ATCC25923, AB459, RN6390 as well as in the *Agr* null strain. The representative genes of interest (*Hla*, *PSMa* and *lukS-PV*) are some of the genes that are very crucial in the pathogenesis of *S. aureus* while causing an infection and hence considered [14,33,34]. The genes expressions were quantified in drug treated and untreated conditions at the late exponential phase of bacterial growth. Briefly, RNA was isolated using Qiagen RNeasy Mini kit as per the kit protocol. cDNA synthesis was carried out using the iScript cDNA synthesis kit. The reaction conditions for cDNA synthesis and qRT-PCR are as follows. Annealing was carried out at 25°C for 5 min, extension at 42°C for 30 min and inactivation of samples at 85°C for 5 min. RT-PCR conditions were initial denaturation at 95°C for 2 min, followed by denaturation (95°C for 15 s), annealing (55.8°C for 15 s) and extension (72°C for 20 s) for 40 cycles. Negative controls containing nuclease-free water instead of cDNA were run in parallel to ensure the samples were free from contamination. Relative gene expression was analyzed using the $2^{-\Delta\Delta\text{CT}}$ method with 16S rRNA as a reference gene [13,35].

2.2.5. Hemolysin assay

S. aureus ATCC25923, *agr* positive laboratory strain RN6390 and AB459 clinical isolate were grown overnight in the presence or absence of the test compounds (diflunisal and bumetanide). Cell free supernatant was taken by filtering the culture supernatant in a $0.22\text{ }\mu\text{m}$ filter and added to $10\text{ }\mu\text{L}$ of sheep erythrocyte cells. 1% Triton X-100 treated erythrocyte cells were taken as positive control. Water with erythrocyte suspension was taken as blank. The hemolytic activity was measured after centrifuging and measuring the OD of the supernatant at 540 nm . Percentage hemolysis was calculated using the formula [(OD of sample – OD of blank / OD of positive control) $\times$ 100] [36].

2.2.6. Biofilm formation by crystal violet assay

In a 96-well microtitre plate, $10\text{ }\mu\text{L}$ of diluted overnight culture ($\sim 1.5 \times 10^7$ cfu/mL) along with varying concentrations of the test compound bumetanide ($0.8\text{--}0.012\text{ }\mu\text{M}$) in $100\text{ }\mu\text{L}$ of TSB was incubated. After 16 h of incubation, biofilm was quantified. Briefly, the loosely bound cells were washed from the wells with sterile $1\times$ PBS and stained

with 0.2% crystal violet. The plates were kept for 20 min, crystal violet discarded and excess stain was washed. The bound dye was eluted with 33% acetic acid and OD measured at 595 nm.

2.3. In vivo studies

2.3.1. Animals

All the animal studies conducted were approved by the Institutional Animal Ethics Committee, SAstra Deemed University (CPCSEA Approval Number: 405/SAstra/IAEC/RPP). The BALB/c mice (*Mus musculus*) were purchased and housed in the Central Animal Facility, SAstra Deemed University. Male immunocompetent mice of 6–8 weeks old, ~25–30 g were caged 5 per group randomly (Table S2). The animals were maintained under standard conditions (individual polypropylene cages, sterilized paddy husk bedding, 12 h light/dark cycle, room temperature of $22 \pm 3^\circ\text{C}$ and 50–70% constant relative humidity). Standard rodent pellet and reverse osmosis drinking water was used for feeding mice. The overall plans of animal studies are provided as a schematic work flow in Fig. S1.

2.3.2. Efficacy of the drugs in a dermonecrosis mice infection model

On day 0, the fur on the flank of the male mice was removed with an electrical trimmer. On day 1, a standardized inoculum of *S. aureus* ($\sim 10^6$ cfu/mL, 50 μL) was injected subcutaneously. Treatment with diflunisal (80 μM) to mice belonging to group V and bumetanide (0.1 μM) to mice belonging to group IV (Table S2) was initiated after 24 h (day 2) and 48 h (day 3) by subcutaneous injection (50 μL) adjacent to the site of bacterial injection. DMSO vehicle (0.3%, 50 μL) was injected to both the mice groups that were infected with *S. aureus* (group III, Table S2) and non-infected (group I, Table S2). The *Agr* null strain (*S. aureus* RN6911) was injected to mice belonging to group II to visually evidence no ulcer development due to virulence factors absence. The ulcer area was measured with a digital Vernier calliper and the weight loss was noted daily. The ulcer area (mm^2) was calculated from the equation (length of the ulcer) $\times$ (width of the ulcer). All the mice were euthanized on day 7 by CO_2 asphyxiation and the ulcer area was excised [14,37,38]. Infected tissue samples were collected and placed in 1 ml of cell culture grade $1 \times \text{PBS}$. The homogenized tissue samples were serially diluted and plated in MSA plates for bacterial cfu enumeration. For histopathology, subjected skin tissue samples were fixed with 10% neutral buffered formalin. Sections were prepared, stained with hematoxylin and eosin to analyze the inflammation. The samples were blindly scored by a veterinary pathologist by numerical scoring guidelines from 0 to 3 [39].

2.3.3. Acute dermal toxicity

Female BALB/c mice of 6–8 weeks old, ~25–30 g were selected and the fur was removed from the dorsal area of the trunk by electrical trimmer. Ten times higher of the effective concentration *i.e.* Diflunisal (800 μM) and bumetanide (1 μM) were prepared in 0.5% hydroxypropyl methyl cellulose (HPMC) and applied to the mice belonging to group III and II respectively (Table S2) ensuring good contact with the skin. A vehicle control mice group (group I, Table S2) is maintained by only using HPMC. Daily feed intake and weekly body weight were measured throughout the observation period. The animals were observed twice every day for a period of 14 days for dermal signs and systemic symptoms like tremors, convulsions, salivation, diarrhoea *etc.*, On the end of 14th day, each of the animal was euthanized by CO_2 asphyxiation and the internal organs were examined for any abnormality [40].

2.3.4. Investigation of microbial tolerance induction/resistance development

Microbial tolerance induction to the test compounds if any was tested as described earlier [14]. Briefly, early exponential phase bacterial culture of *S. aureus* ($\sim 10^6$ cfu/mL, 50 μL) was first injected into the flank of male BALB/c mice. Concurrently, antibiotics erythromycin (8

$\mu\text{g/mL}$) and doxycycline (0.25 $\mu\text{g/mL}$) were injected to group I (Table S2). Diflunisal (80 μM) and bumetanide (0.1 μM) was injected to group III and II respectively (Table S2). After 24 h, weight loss of the animal was noted and the abscess area was excised from the euthanized mice. The excised tissue was bead beaten in cell culture grade $1 \times \text{PBS}$ and plated onto MSA plates. After the incubation period (16 h) a single colony was picked and suspended in $1 \times \text{PBS}$ to get a cell density of $\sim 10^6$ cfu/mL. These bacteria were again injected concurrently with antibiotics or drugs into the flank of the next mouse as mentioned above. This was repeated consecutively till the 10th mice. After the tenth cycle, following euthanization and plating in agar medium, the bacteria were collected and stored as glycerol stock at -80°C . The *in vivo* passaged bacteria with antibiotics or antivirulence drugs along with the non-passaged bacteria were tested for tolerance induction/resistance development if any. Briefly, *S. aureus* was incubated with doxycycline, diflunisal or bumetanide at varying concentrations. CfU enumeration in agar plate was carried out for cells treated with antibiotics and *RNAIII* expression was determined by qRT-PCR for cells that were treated with diflunisal or bumetanide.

2.4. Statistical analyses

All the *in vitro* assays were carried out in triplicates and expressed as mean $\pm$ standard deviation (SD). *In vivo* assays were carried out in groups ($n = 5$). All the statistical analyses were performed with suitable measures in GraphPad Prism 8.0 and considered significant if $p < 0.05$.

3. Results

3.1. Docking to C-LytTR domain of AgrA

The energy minimized reference AgrA inhibitor ligands submitted to PharmaGist server generated many pharmacophores (Table S3). As expected, the resulted ligands from savirin based pharmacophore, when docked, formed H-bond to the highly conserved Tyr-229 of *S. aureus* AgrA and fits into the hydrophobic pocket of AgrA. Of the several hits, four drug molecules were chosen based on the commercial availability. The interaction parameters of the selected drugs are shown in Table 1. The binding pose and interaction plot of bumetanide with maximum binding free energy (-6.3 kcal/mol) is shown in Fig. 1 and other ligands are shown in Figs. S2–S4.

3.2. AgrA inhibition determined by *agrP3::blaZ* reporter assay

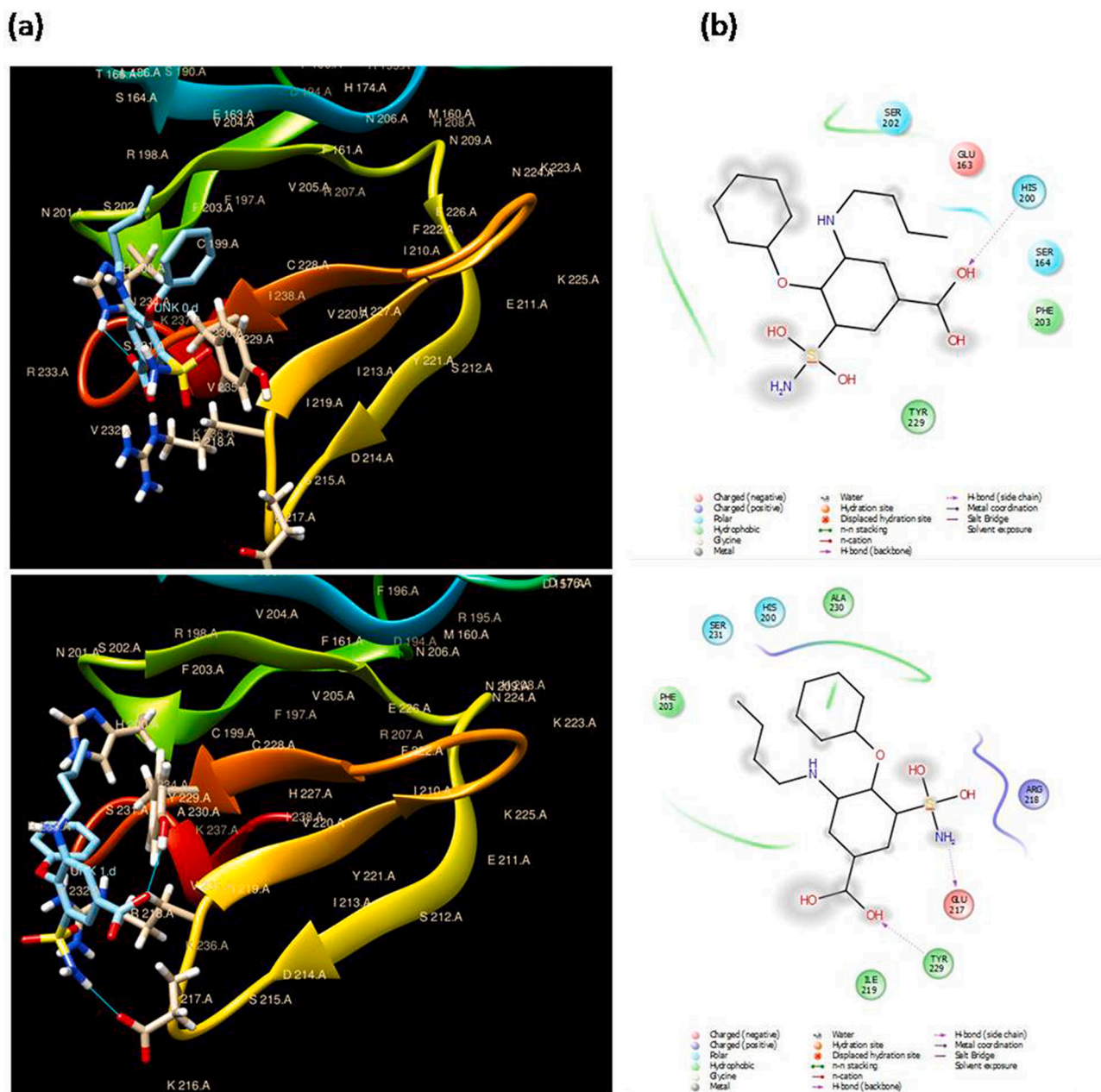
Axitinib exhibited a bimodal action at two different concentrations in a concentration dependent manner. The other tested compounds such as bumetanide dantrolene and diflunisal showed significant AgrA inhibition in a dose dependent manner (Fig. 2). The reference compound, diflunisal showed a maximum inhibition of 53% ($p \leq 0.0001$) at a concentration of 100 μM . It is notable that bumetanide which had maximum binding free energy (-6.3 kcal/mol) *in silico* showed a maximum AgrA inhibition of 70% ($p \leq 0.0001$) at a lesser concentration of 0.1 μM and hence selected for further studies.

3.3. Downregulation of representative virulence genes

Downregulation of the QS virulence genes was tested by qRT-PCR. Bumetanide and diflunisal showed significant downregulation at their tested concentrations (Fig. 3). Bumetanide showed significant reductions ($p \leq 0.0001$) of *RNAIII* (~ 5 log₂-fold), *Hla* (~ 5 log₂-fold), *PSMA* (~ 4 log₂-fold) and *lukS-PV* (~ 3 log₂-fold) (Fig. 3a). Diflunisal showed significant reductions ($p \leq 0.0001$) of *RNAIII* (~ 3 log₂-fold), *Hla* (~ 5 log₂-fold), *PSMA* (~ 4 log₂-fold) and *lukS-PV* (~ 3 log₂-fold) (Fig. 3b). Similarly, significant reduction ($p \leq 0.0001$ to $p \leq 0.01$) in the virulence genes was observed in *S. aureus* ATCC25923 and *Agr* positive laboratory strain RN6390 when treated with bumetanide (Fig. S5).

Docking interaction parameters of the selected drug molecules.

S. no	Ligand number	Drug	Molecular formula	Binding free energy (kcal/mol)	H-bond interacting amino acids	Hydrophobic interactions
1	ZINC11616882	Axitinib	C ₂₂ H ₁₈ N ₄ O ₅	-5.8	His200, Glu217, Tyr229,	Phe161, Phe203, Tyr229,
2	ZINC03813061	Bumetanide	C ₁₇ H ₂₀ N ₂ O ₅ S	-6.3	His200, Glu217, Tyr229,	Phe203, Ile219, Tyr229, Ala230
3	ZINC02568036	Dantrolene sodium	C ₁₄ H ₉ N ₄ NaO ₅	-4.3	His200, His227, Tyr229	Tyr229
4	ZINC00020243	Diflunisal	C ₁₃ H ₈ F ₂ O ₃	-5.6	His200, Arg218	Phe203, Tyr229, Ala230



Importantly, the drugs did not have any effect on the *Agr* null strain, *S. aureus* RN6911 ($\Delta agr::tet(M)$). This observation marks the specificity of the tested drugs and the possible mode of action through AgrA. It was also noted that a significant reduction ($p \leq 0.0001$) in the hemolytic activity in drug treated conditions compared to untreated samples through hemolysin assay (Fig. S6).

Inhibition of Agr is expected to increase the biofilm formation. Significant increase ($p \leq 0.0001$ to $p \leq 0.05$) in biofilm formation was seen in *S. aureus* ATCC and clinical strains when treated with bumetanide at different concentrations (Fig. 4). Naturally, the *Agr* null strain, *S. aureus* RN6911 ($\Delta agr::tet(M)$) is known for good biofilm formation because of the dysfunctional *Agr*. In our study also, the *Agr* null mutant formed

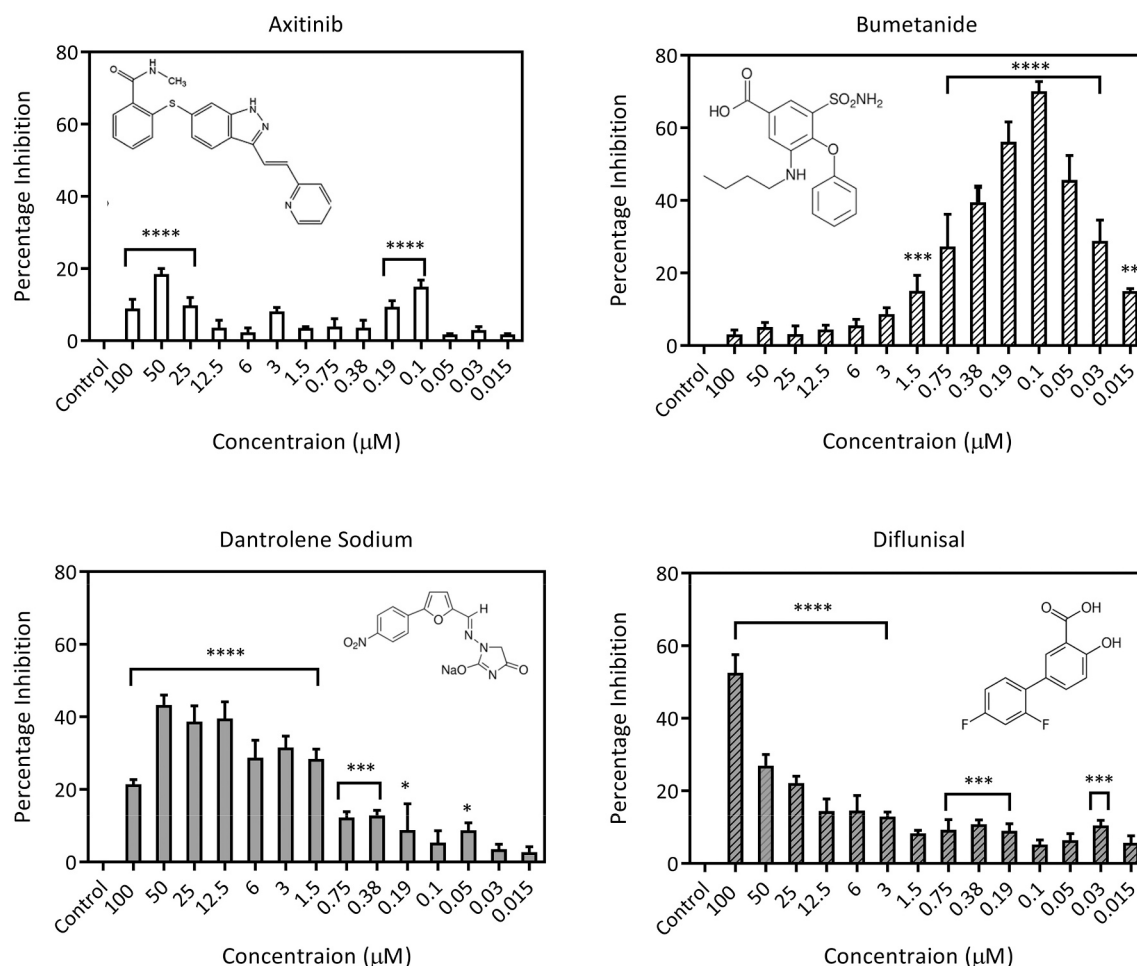


Fig. 2. Percentage inhibition of AgrA using the selected test compounds by *agrP3::blaZ* reporter assay. All the assays were performed in triplicates. One-way ANOVA and Dunnett's multiple comparisons test was conducted and considered significant, **** $p \leq 0.0001$; *** $p \leq 0.001$; ** $p \leq 0.01$; * $p \leq 0.05$.

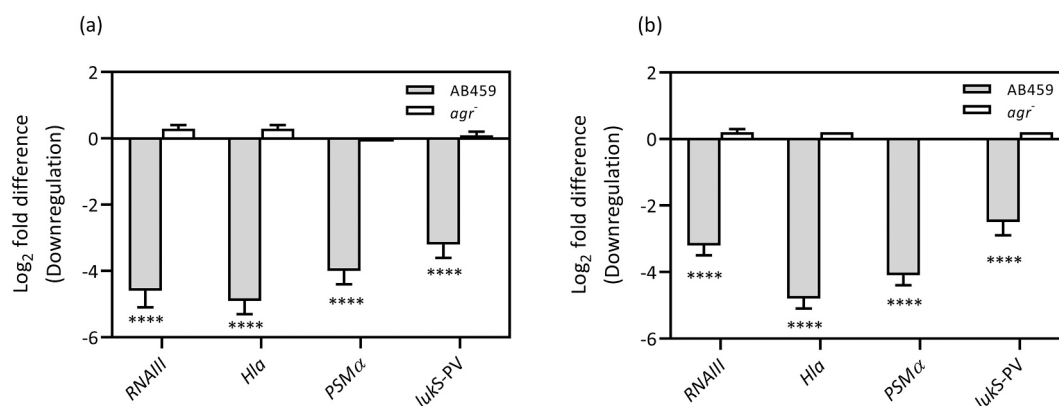


Fig. 3. Downregulation of virulence genes expression expressed as log₂ fold difference. *S. aureus* clinical isolate AB459 and *S. aureus* Agr null strain (RN6911) gene expression when treated with (a) bumetanide and (b) diflunisal. 16S rRNA was used as reference gene for normalization. Data represents the average of 3 individual experiments done in triplicates. Significant downregulation was noted in all the virulence genes tested (**** $p \leq 0.0001$).

good biofilm but no significant increase was noted when treated with bumetanide. It is an expected result as the Agr null strain does not have a functional Agr for the compounds to act upon and contribute further biofilm increase. This indirectly suggests that bumetanide acts through Agr QS system.

3.5. In vivo – drug efficacy

The efficacy of bumetanide and diflunisal were tested in an *in vivo* dermonecrosis mice model and the results were shown in Fig. 5. Fig. 5a, shows the ulcer developed on day 3 post infection on different groups where ulcer development was significantly reduced in the drug treated mice. Also, it is clearly seen that ulcer development occurred until day 3 post infection and subsequently wound healing started with a reduction

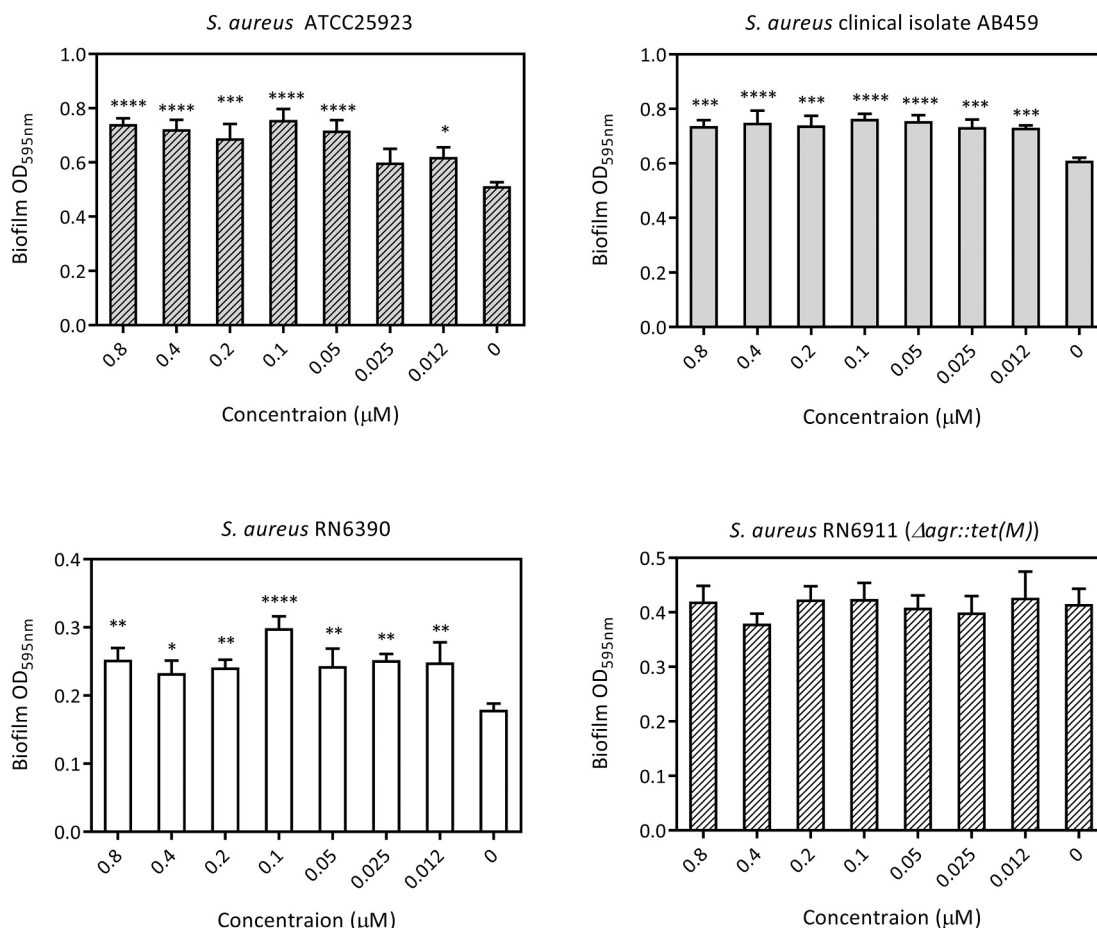


Fig. 4. Biofilm formation by the strains of *S. aureus* in the absence and presence of bumetanide at varying concentrations. All the assays were performed in triplicates. One-way ANOVA and Dunnett's multiple comparisons test was conducted and considered significant, ****p ≤ 0.0001; ***p ≤ 0.001; **p ≤ 0.01; *p ≤ 0.05.

in the ulcer size (Fig. 5b). It is quite notable that even during the development of ulcer till day 3, the treatment groups showed less ulcer development compared to vehicle treatment alone in the infected mice. Both the reference and the test compounds showed significant ulcer inhibition but bumetanide treatment was found to be the most effective ($p \leq 0.01$) when compared to others. Fig. 5c, shows significantly less ($p \leq 0.0001$) bacterial load in treatment groups compared to the infected group treated with only vehicle. Weight loss measurements (Fig. S7) showed that body weight loss was more in the case of infected mice treated with vehicle (0.3% DMSO) when compared to the drug treatment groups. After day 4, no weight loss was seen and could be speculated due to the recovery of animals by drug treatment as well as by the probable boosting of host immunity. Moreover, the spleen weight measured after euthanization on day 7 showed significant increase ($p \leq 0.05$) in the infected group compared to vehicle control which is because of the increased inflammation. Whereas, the spleen weights were found normal in the treatment groups (Fig. S8).

3.6. Histopathology and acute dermal toxicity

Histology sections showed marked differences between the infected and the treatment groups in the inflammatory scores (Fig. 6). High scores indicate more inflammation whereas low scores indicate less or no inflammation. The uninfected control group showed normal tissue (Fig. 6a) without apparent changes (Score, 0) which is equivalent with the *Agr* null mutant strain where focal/very minimal inflammatory cell infiltrate (Score, 0) (Fig. 6b). Infected mice with *S. aureus* AB459 showed marked acute inflammation (Fig. 6c) often involving deep skeletal

muscle, necrosis, and edema with minimal chronic changes (Score, 3). On the other side, infected mice treated with bumetanide, 0.1 μM showed minimal acute inflammation, necrosis and edema, and presence of chronic changes including chronic inflammation, fibroblast proliferation, granulation tissue and fibrosis (Score, 1) (Fig. 6d). Similarly, diflunisal treatment with 80 μM (Fig. 6e) resulted in moderate acute inflammation, necrosis and edema with minimal chronic changes (Score, 2).

Acute dermal toxicity tested showed no systemic symptoms or abnormal observations in the internal organs in all the groups. Also, the feed intake and body weight measured did not have any difference between the test substances and vehicle groups indicating no dermal toxicity (Table S4).

3.7. In vivo resistance

Prolonged antibiotic usage may lead to induce tolerance/resistance development. Hence, it becomes necessary to validate any new compound for tolerance induction to the targeted bacterial strains. In the present study, continuous exposure of the clinical isolate *S. aureus* AB459 to the antibiotic doxycycline has induced more tolerance ($p \leq 0.01$) that can be inferred from the increased survival of the exposed bacterial cells (Fig. 7a). If the cells had acquired tolerance to the tested drugs a reduction in the fold change of the QS regulator *RNAIII* is expected. Whereas the tested drugs bumetanide and diflunisal in the present study, did not show any significant change ($p \geq 0.05$) in *RNAIII* expression between the non-passaged and passaged cells (Fig. 7b and c). This indicates the test substances did not attain tolerance on repeated

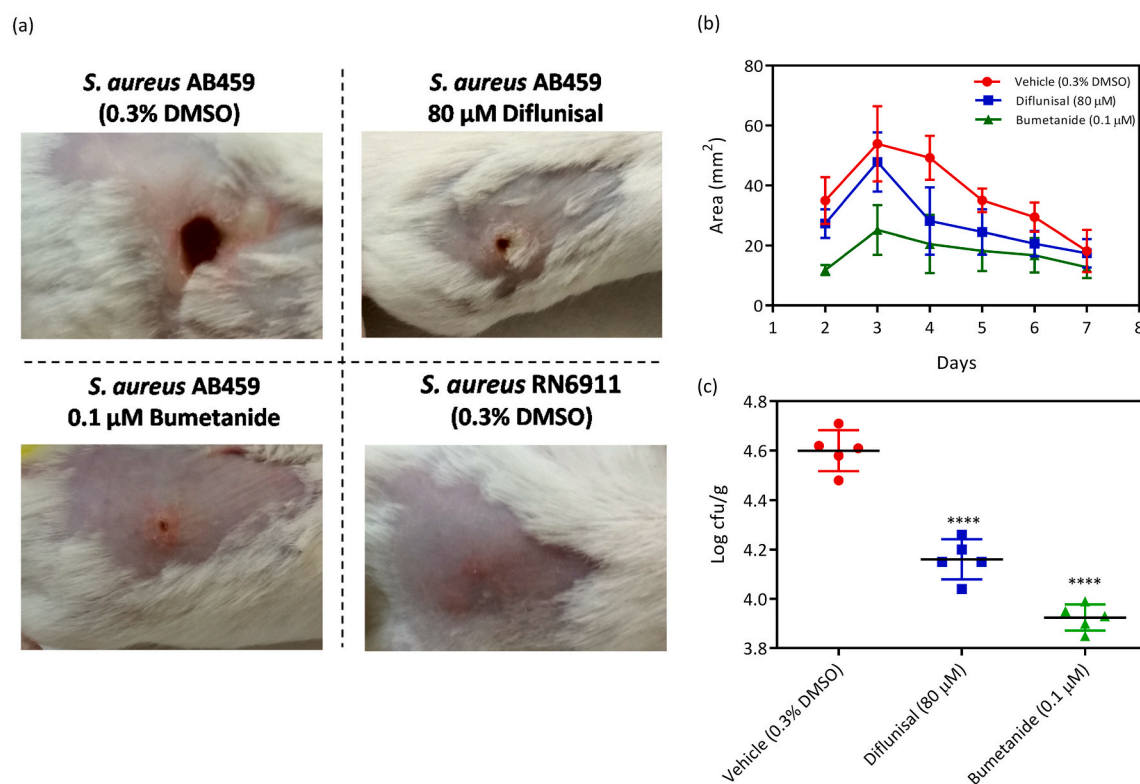


Fig. 5. (a) *In vivo* dermonecrosis mice model infected with clinical isolate *S. aureus* AB459 showing the efficacy of diflunisal (80 µM) and bumetanide (0.1 µM) compared with vehicle (0.3% DMSO) on day 3. Compounds were administered after 24 and 48 h in *S. aureus* AB459 infected mice. Visual wound healing was observed and comparable to *S. aureus* *Agr* null strain (*S. aureus* RN6911) injected mice (no ulcer development) (b) Ulcer area calculated on untreated and treated groups at different days. Ulcer area was significantly less for diflunisal (filled squares, $p \leq 0.05$) and bumetanide (filled triangles, $p \leq 0.01$) compared to vehicle (filled circles) by an unpaired *t*-test. (c) Cfu enumerated from the ulcer tissue samples of *S. aureus* AB459 infected/treated mice on day 7 after euthanization. Cfu was significantly less in the case of treated groups (by an unpaired *t*-test with $n = 5$ and **** $p \leq 0.0001$) when compared to the untreated group.

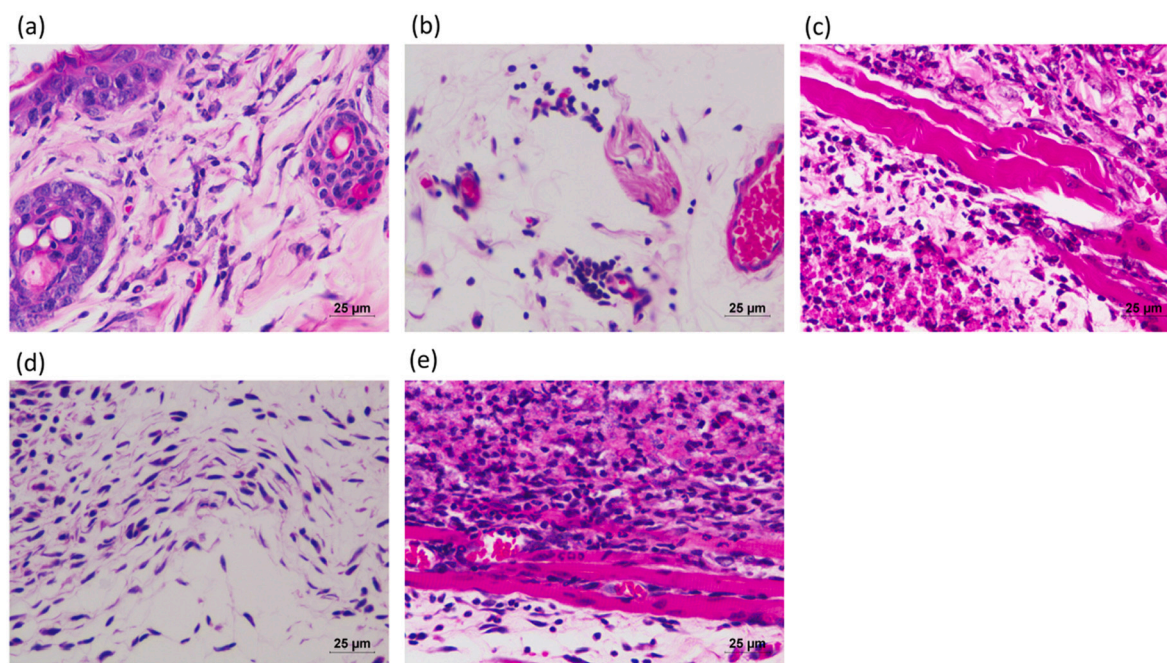


Fig. 6. Histopathological analysis of tissue samples. (a) Control group administered with vehicle (0.3% DMSO). (b) Group infected with Δagr strain (*S. aureus* RN6911) and administered with 0.3% DMSO. (c) Group infected with *S. aureus* AB459 and administered with 0.3% DMSO. (d) Group infected with *S. aureus* AB459 and treated with bumetanide, 0.1 µM. (e) Group infected with *S. aureus* AB459 and treated with diflunisal, 80 µM.

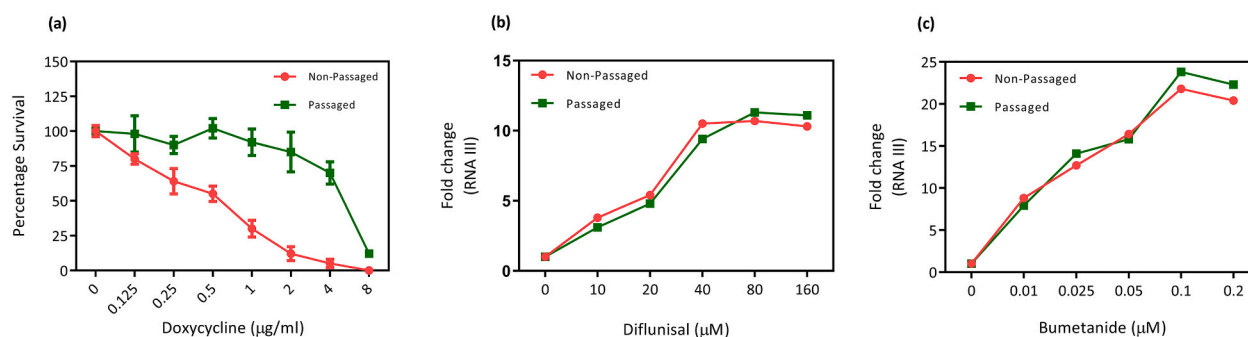


Fig. 7. *In vivo* passaged (filled squares) and non-passaged (filled circles) clinical *S. aureus* exposed to (a) doxycycline (b) diflunisal and (c) bumetanide at varying concentrations. Survival percentage increased in cells that were passaged with doxycycline compared to non-passaged cells ($p \leq 0.01$ by an unpaired t-test). In the case of test compound (bumetanide) and reference compound (diflunisal) treatments, no change was observed between passaged and non-passaged cells (ns, $p \geq 0.05$ by an unpaired t-test).

Table 2
Original and repurposed activities of the selected FDA approved drugs.

S. no	FDA approved drug	Original use	Newly reported activity relevant to Agr	Reference
1	Axitinib	Anticancer and antitumor	AgrA _C binding - <i>in silico</i> ; AgrA inhibition - <i>in vitro</i>	This study
2	Bumetanide	Diuretic	AgrA _C binding - <i>in silico</i> ; AgrA inhibition - <i>in vitro</i> Antivirulence - <i>in vitro</i> and <i>in vivo</i>	This study
3	Dantrolene Sodium	Muscle relaxant	AgrA _C binding - <i>in silico</i> ; AgrA inhibition - <i>in vitro</i>	This study
4	Diflunisal	Analgesic and anti-inflammatory	Agr mediated antivirulence AgrA _C binding - <i>in silico</i> ; AgrA inhibition - <i>in vitro</i> Antivirulence - <i>in vitro</i> and <i>in vivo</i>	[31,32] This study

exposures proving it to be a highly desirable feature for therapeutic purposes (Table 2).

4. Discussion

In *S. aureus*, Agr QS system is responsible for the pathogenicity of the microbe towards the host system by secretion of various virulence factors. Several studies have highlighted the importance of targeting Agr QS system in tackling the MDR problem in *S. aureus* [7,14,41]. Of all the Agr operon products, AgrA regulator is considered as a potential drug target due to its unique small and soluble protein nature [19,31]. In the present study, we have screened FDA approved drugs that possess AgrA_C binding through molecular docking studies. All the selected compounds (Table 1) were capable of binding to the conserved amino acid residue, Tyr229 as well as interacting to the hydrophobic residues in the active site of AgrA. This rewards the choice of pharmacophore based search especially savirin molecule that originally binds to the conserved amino acid Tyr229 [14]. A previous study which reported antivirulence activity of diflunisal has speculated it to have a possible binding to the C-terminal domain of AgrA [31]. In our present study, we have confirmed their speculation through docking studies and considered diflunisal as a reference compound in all the assays. Binding of bumetanide to AgrA_C especially to the conserved amino acid residue Tyr-229 with high binding free energy is remarkable (Fig. 1) as Tyr-229 is involved in the maximal activation of AgrA transcription [24]. Other amino acid residues adjacent to Tyr-229 have also been shown important. For instance, defective Agr function in clinical *S. aureus* isolates has been described when Arg-218 is mutated [42]. Further, through *agrP3::blaZ* reporter assay the AgrA inhibitory roles of the four tested compounds were

proven in a dose dependent manner (Fig. 2). Only axitinib exhibited bimodal action and demands further research as it is uncertain how the two different concentration will influence the host in practical applications. The Agr inhibitory activity was an expected one as all the compounds were resulted from a savirin based pharmacophore. Savirin is one of the successful molecules with reported antivirulence activity in mouse models [14].

Curtailling the virulence of pathogenic *S. aureus* make them less likely to colonize the host tissues and boost the immune system [14,33]. The virulence factors are secreted in the post-exponential phase that helps in invading the host tissues by tissue penetration and cell lysis [33]. The significant fold reductions of *RNAIII* and other virulence factors in our present study are worth mentioning (Fig. 3). Downregulation of these virulence genes indicates the potent antivirulence activity of the tested compounds and suggests QS target specific action by the following observations; (i) significant downregulation of the virulence genes especially, *PSMA* that usually gets triggered an upregulated expression when bound to AgrA independent of *RNAIII* [43]. (ii) Compounds treated on the null Agr, *S. aureus* RN6911 (*Δagr::tet(M)*) did not exhibit any fold change. Hemolysins are the most characterized virulence factor in *S. aureus* that involve in pore formation on the cell membrane. Treatment with bumetanide has resulted significantly reduced hemolytic activity which suggest the less expression of hemolysin toxin (Fig. S6).

Dysfunctional Agr results in decreased virulence but, increased biofilm formation *in vitro* [15–17,21]. This could be an evidence for target specific action by Agr inhibitors. Bumetanide increased the biofilm formation in polystyrene surface of microtitre plates (Fig. 4). Our observed results are in corroboration with the earlier studies and provide indirect evidence that our test substances are acting through Agr mediated QS. However, this increased biofilm forming behaviour is a limitation as it promotes the biofilm and makes clinical therapy challenging [21]. We agree with this viewpoint where the increased biofilm formation might act as an antibiotic diffusion barrier. Despite this constraint, reduced virulence is more significant to control the pathogenicity of bacteria in host and targeting down AgrA becomes even more significant. Hence, it necessitates the sensible use of these test substances in combination with adjunctive therapies. On the other side, instead of choosing implant-associated infection or established chronic disease focussing on other relevant infections with these types of inhibitors will be judicious [32]. For instance, in the present study bumetanide is proven to be more effective in a dermonecrosis model treating skin infection (Fig. 5). So, the choice of disease selection for therapy with these types of compounds is highly advisable. Despite these challenges, significant downregulation of AgrA (Fig. 2), virulence genes (Fig. 3) and effective control of ulcer development in mouse dermonecrosis model (Fig. 5) by these substances are impressive for adjunctive therapies with antibiotics.

Survival stress on the pathogen plays an important role in phenotypic

switching from wild type to tolerant and resistant bacteria [44]. Bumetanide did not show any killing effect on the bacterial strains till a tested concentration of 100 μ M (Table S5). However in another study, bumetanide has been shown to possess antimicrobial activity at very high concentrations of >100 μ g/ml [45] which is equivalent to >275 μ M. The antimicrobial tested concentration of our study is 1000 times higher than the AgrA inhibitory concentration (i.e. 0.1 μ M). In that concentration also we have not observed any bactericidal effect. Hence, it confirms the observed activities are not through bacteriostatic or bactericidal effects. Therefore, our results positively contribute to the elimination of survival stress as well and reduce the chance of drug resistance development.

Our *in vivo* efficacy results (Fig. 5) were in corroboration with an earlier study where similar results have been shown in Balb/C mice infection model with an AgrA dependent QS inhibitor molecule called apicidin [46]. In the present study, diflunisal and bumetanide treated mice groups exhibited anti-inflammatory profiles. Especially bumetanide, showed minimal inflammation with fibroblast proliferation (Fig. 6d) indicating wound healing [39]. Further, no dermal toxicity favours bumetanide to be a good option for SSTI topical drug formulations research.

It is highly important to predict the tolerance induction properties for a new test substance on repeated exposures. Though, it is currently not possible to predict this in direct human infections, we have passaged the MDR clinical isolate to our test compound over a period on tube cultures as well as on a mice infection model. The *in vitro* results were in concordance with the *in vivo* results. The test results suggest that the compounds did not induce any drug tolerance or resistance for the tested concentration and time period in *S. aureus* as conventional antibiotics did (Fig. 7). However, these preliminary data require further studies at different exposure concentrations and time period with more clinical isolates to make a more decisive conclusion.

It is quite interesting to note that, a study previously has shown bumetanide as well as a few other diuretics are capable of enhancing the disease resistance in *Arabidopsis* seedlings priming the plant immunity [47]. Such as role in mammalian system is uncertain at this stage but would be highly appreciable in drug development research if a possibility exists. Another study discussed the use of bumetanide and other diuretic drugs in treating extracellular fluid volume expansion and edema [48]. Such activity might be useful in the drug formulation research for skin infections that are often associated with edema. Nevertheless, in the present study, we have reported for the first time, the potential antivirulence activity of bumetanide acting through AgrA mediated QS inhibition.

5. Conclusions

In the present study, the diuretic drug bumetanide was successfully repurposed to show significant antivirulence properties through AgrA QS inhibition in *S. aureus*. Though, the present study holds limitation in showing bumetanide's direct binding between the purified protein AgrA and DNA, the obtained results of bumetanide are highly substantial with the reference antivirulence compound, diflunisal. Bumetanide showed many-fold reduction of the virulence genes at a very lesser concentration (0.1 μ M). In the dermonecrosis model, bumetanide proved to be highly effective in controlling the infection without any tolerance/resistance induction. Also, no modulation in virulence in the Agr null strain suggests that the reported compounds likely exhibit target-specific action. However, further studies describing the synergistic role of antivirulence agents like bumetanide and conventional antibiotics are requisite for addressing clinical infections effectively.

CRediT authorship contribution statement

P Balamurugan: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - Original draft, Writing - Review

& editing, Project administration, Funding acquisition. **Solomon Adline Princy:** Writing - Review & editing, Supervision. **C David Raj:** Investigation, Writing - Review & editing, Supervision.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

Acknowledgements

We would like to thank: the management of SASTRA Deemed University for providing the required infrastructure to complete the research work; Dr M. N. Sumana (Professor and Head of Department, Department of Microbiology, JSS Medical College, Mysore, India) for providing the clinical isolate (*S. aureus* AB459); Professor Christiane Wolz (APL Professor, Department of Medical Microbiology and Hygiene at the Faculty of Medicine of the University of Tübingen) for the Agr positive, Agr null *S. aureus* and reporter strains.

Funding

This work was supported by the Science and Engineering Research Board, India [Grant No. PDF/2015/000668].

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lfs.2021.119306>.

References

- [1] E.B. Chahine, A.J. Sucher, Skin and soft tissue infections, *PSAP* 1 (2015) 1–27.
- [2] M.D. King, B.J. Humphrey, Y.F. Wang, E.V. Kourbatova, S.M. Ray, H.M. Blumberg, Emergence of community-acquired methicillin-resistant *Staphylococcus aureus* USA 300 clone as the predominant cause of skin and soft-tissue infections, *Ann. Intern. Med.* 144 (5) (2006) 309–317.
- [3] G. Lazarus, J. Audrey, Comprehensive assessment of risk factors associated with methicillin-resistant *Staphylococcus aureus* (MRSA) infection in Asia: a systematic review, *J. Asian Med. Stud. Assoc.* 7 (1) (2019).
- [4] D.L. Stevens, A.L. Bisno, H.F. Chambers, E.P. Dellinger, E.J. Goldstein, S. L. Gorbach, et al., Practice guidelines for the diagnosis and management of skin and soft tissue infections: 2014 update by the Infectious Diseases Society of America, *Clin. Infect. Dis.* 59 (2) (2014) e10–e52.
- [5] T.J. Foster, Antibiotic resistance in *Staphylococcus aureus*. Current status and future prospects, *FEMS Microbiol. Rev.* 41 (3) (2017) 430–449.
- [6] A.O. Shittu, M. Kaba, S.M. Abdulgader, Y.O. Ajao, M.O. Abiola, A.O. Olatimehin, Mupirocin-resistant *Staphylococcus aureus* in Africa: a systematic review and meta-analysis, *Antimicrob. Resist. Infect. Control* 7 (1) (2018), 101.
- [7] M.I. Abd El-Hamid, E.Y. El-Naenaeey, T.M. Kandeel, W.A.H. Hegazy, R.A. Mosbah, M.S. Nassar, M.A. Bakhrebah, W.H. Abdulaal, N.A. Alhakamy, M.M. Bendary, Promising antibiofilm agents: recent breakthrough against biofilm producing methicillin-resistant *Staphylococcus aureus*, *Antibiotics* 9 (2020) 667.
- [8] W.A. Hegazy, M.T. Khayat, T.S. Ibrahim, M.S. Nassar, M.A. Bakhrebah, W. H. Abdulaal, N.A. Alhakamy, M.M. Bendary, Repurposing anti-diabetic drugs to cripple quorum sensing in *Pseudomonas aeruginosa*, *Microorganisms* 8 (2020) 1285.
- [9] G.J. Lyon, P. Mayville, T.W. Muir, R.P. Novick, Rational design of a global inhibitor of the virulence response in *Staphylococcus aureus*, based in part on localization of the site of inhibition to the receptor-histidine kinase, AgrC, *Proc. Natl. Acad. Sci. U. S. A.* 97 (24) (2000) 13330–13335.
- [10] N. Balaban, A. Giacometti, O. Cirioni, Y. Gov, R. Ghiselli, F. Mocchegiani, et al., Use of the quorum-sensing inhibitor RNAIII-inhibiting peptide to prevent biofilm formation in vivo by drug-resistant *Staphylococcus epidermidis*, *J. Infect. Dis.* 187 (4) (2003) 625–630.
- [11] G. Dell'Acqua, A. Giacometti, O. Cirioni, R. Ghiselli, V. Saba, G. Scalise, et al., Suppression of drug-resistant staphylococcal infections by the quorum-sensing inhibitor RNAIII-inhibiting peptide, *J. Infect. Dis.* 190 (2) (2004) 318–320.
- [12] T.B. Rasmussen, T. Bjarnsholt, M.E. Skindersoe, M. Hentzer, P. Kristoffersen, M. Kôte, et al., Screening for quorum-sensing inhibitors (QSI) by use of a novel genetic system, the QSI selector, *J. Bacteriol.* 187 (5) (2005) 1799–1814.
- [13] P. Balamurugan, V. Praveen Krishna, D. Bharath, R. Lavanya, P. Vairaprakash, S. Adline Princy, *Staphylococcus aureus* quorum regulator SarA targeted compound, 2-[(Methylamino) methyl] phenol inhibits biofilm and down-regulates virulence genes, *Front. Microbiol.* 8 (2017) 1290.
- [14] E.K. Sully, N. Malachowa, B.O. Elmore, S.M. Alexander, J.K. Fleming, B.M. Gray, et al., Selective chemical inhibition of agr quorum sensing in *Staphylococcus aureus*

- promotes host defense with minimal impact on resistance, *PLoS Pathog.* 10 (2014), e1004174.
- [15] C. Vuong, H.L. Saenz, F. Götz, M. Otto, Impact of the agr quorum-sensing system on adherence to polystyrene in *Staphylococcus aureus*, *J. Infect. Dis.* 182 (6) (2000) 1688–1693.
 - [16] B.R. Boles, A.R. Horswill, Agr-mediated dispersal of *Staphylococcus aureus* biofilms, *PLoS Pathog.* 4 (4) (2008), e1000052.
 - [17] S.S. Dastgheyb, et al., Role of phenol-soluble modulins in formation of *Staphylococcus aureus* biofilms in synovial fluid, *Infect. Immun.* 83 (2015) 2966–2975.
 - [18] C. Jenul, A.R. Horswill, Regulation of *Staphylococcus aureus* virulence, *Gram Posit. Pathog.* (2019 Oct 1) 669–686.
 - [19] M. Greenberg, D. Kuo, E. Jankowsky, L. Long, C. Hager, K. Bandi, et al., Small-molecule AgrA inhibitors F12 and F19 act as antivirulence agents against Gram-positive pathogens, *Sci. Rep.* 8 (1) (2018) 1–2.
 - [20] A.M. Salam, C.L. Quave, Targeting virulence in *Staphylococcus aureus* by chemical inhibition of the accessory gene regulator system *in vivo*, *M Sphere* 3 (1) (2018).
 - [21] I.F. Bezdar, A.A. Mashruwala, J.M. Boyd, et al., Drug-like fragments inhibit agr-mediated virulence expression in *Staphylococcus aureus*, *Sci. Rep.* 9 (2019), 6786, <https://doi.org/10.1038/s41598-019-42853-z>.
 - [22] S. Das, A. Dasgupta, S. Chopra, Drug repurposing: a new front in the war against *Staphylococcus aureus*, *Future Microbiol.* (8) (2016) 1091–1099.
 - [23] M. Gajdacs, G. Spengler, The role of drug repurposing in the development of novel antimicrobial drugs: non-antibiotic pharmacological agents as quorum sensing-inhibitors, *Antibiotics (Basel)* 8 (4) (2019 Dec 17) 270, <https://doi.org/10.3390/antibiotics8040270>. PMID: 31861228; PMCID: PMC6963710.
 - [24] S.S. Nicod, R.O. Weinzierl, L. Burchell, A. Escalera-Maurer, E.H. James, S. Wigneshweraraj, Systematic mutational analysis of the LytR DNA binding domain of *Staphylococcus aureus* virulence gene transcription factor AgrA, *Nucleic Acids Res.* 42 (20) (2014) 12523–12536.
 - [25] R.J. Gordon, F.D. Lowy, Pathogenesis of methicillin-resistant *Staphylococcus aureus* infection, *Clin. Infect. Dis.* 46 (Suppl. 5) (2008) S350–S359.
 - [26] P.G. Leonard, I.F. Bezdar, D.J. Sidote, A.M. Stock, Identification of a hydrophobic cleft in the LytR domain of AgrA as a locus for small molecule interactions that inhibit DNA binding, *Biochemistry* 51 (50) (2012) 10035–10043.
 - [27] R. Agrawal, P. Jain, S.N. Dikshit, R.S. Bahare, S. Ganguly, Ligand-based pharmacophore detection, screening of potential pharmacophore and docking studies, to get effective glycogen synthase kinase inhibitors, *Med. Chem. Res.* 22 (11) (2013) 5504–5535.
 - [28] G.O. Ambrose, O.J. Afees, N.C. Nwamaka, N. Simon, A.A. Oluwaseun, T. Soyinka, et al., Selection of Luteolin as a potential antagonist from molecular docking analysis of EGFR mutant, *Bioinformation* 14 (5) (2018) 241.
 - [29] K.S. Gurusinge, A. Mishra, S. Mishra, Glucose-regulated protein 78 substrate-binding domain alters its conformation upon EGCG inhibitor binding to nucleotide-binding domain: molecular dynamics studies, *Sci. Rep.* 8 (1) (2018) 1–11.
 - [30] V. Shukla, Z. Bhathena, Broad spectrum anti-quorum sensing activity of tannin-rich crude extracts of Indian medicinal plants, *Scientifica* (2016).
 - [31] V. Khodavardian, M. Pesho, B. Truitt, L. Bollinger, P. Patel, S. Nithianantham, et al., Discovery of antivirulence agents against methicillin-resistant *Staphylococcus aureus*, *Antimicrob. Agents Chemother.* 57 (8) (2013) 3645–3652.
 - [32] A.S. Hendrix, T.J. Spoonmore, A.D. Wilde, N.E. Putnam, N.D. Hammer, D. J. Snyder, et al., Repurposing the nonsteroidal anti-inflammatory drug diflunisal as an osteoprotective, antivirulence therapy for *Staphylococcus aureus* osteomyelitis, *Antimicrob. Agents Chemother.* 60 (9) (2016) 5322–5330.
 - [33] C. Kong, H.M. Neoh, S. Nathan, Targeting *Staphylococcus aureus* toxins: a potential form of anti-virulence therapy, *Toxins* 8 (3) (2016) 72.
 - [34] D. Oliveira, A. Borges, M. Simões, *Staphylococcus aureus* toxins and their molecular activity in infectious diseases, *Toxins* 10 (6) (2018) 252.
 - [35] S. Vasudevan, G. Thamil Selvan, S. Bhaskaran, N. Hari, A.P. Solomon, Reciprocal cooperation of type A procyanidin and nitrofurantoin against multi-drug resistant (MDR) UPEC: a pH-dependent study, *Front. Cell. Infect. Microbiol.* 10 (2020 Aug 11) 421, <https://doi.org/10.3389/fcimb.2020.00421>. PMID: 32850505; PMCID: PMC7431559.
 - [36] M. Hema, J. Saikiran, C. Kamleswar, V. Sridharan, G. Kaur, P. Balamurugan, M. N. Sumana, A. Princy, Design, synthesis and validation of cqs receptor agonist to modulate the quorum sensing circuit of *Vibrio cholerae*—a molecular mimic therapy, *Int J Pharm Pharm Sci* 7 (13) (2015).
 - [37] N. Malachowa, S.D. Kobayashi, K.R. Braughton, F.R. DeLeo, Mouse model of *Staphylococcus aureus* skin infection, in: I.C. Allen (Ed.), *Mouse Models of Innate Immunity*, Springer Protocols, NJ, 2013, pp. 109–116.
 - [38] D.A. Todd, C.P. Parlet, H.A. Crosby, C.L. Malone, K.P. Heilmann, A.R. Horswill, et al., Signal biosynthesis inhibition with ambuic acid as a strategy to target antibiotic-resistant infections, *Antimicrob. Agents Chemother.* 61 (8) (2017), e00263-17.
 - [39] C.W. Tseng, J.C. Biancotti, B.L. Berg, D. Gate, S.L. Kolar, S. Müller, et al., Increased susceptibility of humanized NSG mice to Pantone-Valentine leukocidin and *Staphylococcus aureus* skin infection, *PLoS Pathog.* 11 (11) (2015), e1005292.
 - [40] OECD, Test no. 402: acute dermal toxicity, in OECD Guidelines for the Testing of Chemicals, Section 4, Health Effects, Organisation for Economic Cooperation and Development, OECD Publishing, Paris, 1987.
 - [41] X. Zhao, Z. Yu, T. Ding, Quorum-sensing regulation of antimicrobial resistance in bacteria, *Microorganisms* 8 (3) (2020 Mar 17) 425, <https://doi.org/10.3390/microorganisms8030425>. PMID: 32192182; PMCID: PMC7143945.
 - [42] B. Shopsin, C. Eaton, G.A. Wasserman, B. Mathema, R.P. Adhikari, S. Agolory, et al., Mutations in agr do not persist in natural populations of methicillin-resistant *Staphylococcus aureus*, *J. Infect. Dis.* 202 (10) (2010) 1593–1599.
 - [43] M. Thoendel, J.S. Kavanaugh, C.E. Flack, A.R. Horswill, Peptide signaling in the staphylococci, *Chem. Rev.* 111 (1) (2011) 117–151.
 - [44] J. Dawan, S. Wei, J. Ahn, Role of antibiotic stress in phenotypic switching to persist cells of antibiotic-resistant *Staphylococcus aureus*, *Ann. Microbiol.* 70 (1) (2020) 1–8.
 - [45] I.M. Salem, S.M. Mostafa, O.I. El-Sabbag, T.S. Ibrahim, A. Abdulrahmanf, M. M. Bendary, Synthesis and antimicrobial evaluation of novel hydrazones and 1, 3, 4-oxadiazoles incorporating bumetanide derivatives, *Rec. Pharm. Biomed. Sci.* 2 (2) (2018) 52–66.
 - [46] C.P. Parlet, J.S. Kavanaugh, H.A. Crosby, H.A. Raja, T. El-Elmat, D.A. Todd, et al., Apicidin attenuates MRSA virulence through quorum-sensing inhibition and enhanced host defense, *Cell Rep.* 27 (1) (2019) 187–198.
 - [47] Y. Noutoshi, M. Ikeda, K. Shirasu, Diuretics prime plant immunity in *Arabidopsis thaliana*, *PLoS One* 7 (10) (2012).
 - [48] D.H. Ellison, Clinical pharmacology in diuretic use, *CJASN* 14 (8) (2019) 1248–1257.